

Amrita Vishwa Vidyapeetham
B.Tech. Degree Examinations – November 2010
First semester

CHY100 Chemistry

(Common to all branches)

Time: Three hours

Maximum: 100 Marks

Answer all questions**PART A****(10 X 2 = 20 Marks)**

1. What are the requirements for the molecules to exhibit liquid crystalline behavior?
2. Predict the types of electronic transitions possible for the following compounds
 - a. $\text{CH}_3\text{-CH}_3$
 - b. $\text{C}_2\text{H}_5\text{OH}$
3. Mention the difference between freezing point and eutectic point.
4. The presence of CO_2 in boiler feed water should be avoided, why?
5. Water used in textile and dyeing industry should be free from Ca and Mg salts, why?
6. Depict the different vibrations observed in CO_2 molecule.
7. Why does corrosion occur in a steel pipe at the steel-copper joint?
8. Differentiate between bottom-up and top-down techniques used in the preparation of nanomaterials.
9. Bring out the relevance of condensed phase rule.
10. Mention the Pilling-Bedworth ratio and what is its significance.

PART B**(5 X 16 = 80 Marks)**

11. A. The standard hard water contains 15g of CaCO_3 per litre. 20ml of this solution consumes 25ml of EDTA solution. 100ml of sample water requires 18ml of the above EDTA solution. The same sample after boiling requires 12ml of EDTA solution. Calculate the permanent, total and temporary hardness in terms of ppm and degree French. (7 marks)
B. What is meant by the term chemical shift in NMR spectroscopy? Explain with example how electronegativity affects the chemical shift value. (5 marks)
C. Explain lyotropic and thermotropic liquid crystals with examples. (4 marks)
12. A. With the help of a neat phase diagram of lead-silver, explain salient features of two component system. (8 marks)
B. Calculate the number of components and degrees of freedom in i. mixture of water (g), hydrogen (g) and oxygen (g) ii. Aqueous solution of acetic acid iii. Saturated solution of sodium chloride in equilibrium with its vapor and iv. $2\text{KClO}_3(\text{s}) \leftrightarrow 2\text{KCl}(\text{s}) + 3\text{O}_2(\text{g})$ (8 marks)
13. A. Explain with a neat sketch, the electro dialysis method for desalination of water. (5 marks)
B. Discuss the applications of nanomaterials. (6 marks)
C. Explain different methods to control corrosion. (5 marks)
14. A. Discuss the mechanism of differential aeration corrosion with example. (5 marks)
B. What is the effect of solvent and auxochrome on wavelength and intensity? (5 marks)
C. How is the corrosion rate determined by weight loss method? (6 marks)
15. A. Mention the various possible molecular vibrations for triatomic molecule with diagram. (6 marks)
B. Discuss the internal conditioning methods for boiler feed water. (4 marks)
C. Explain the mechanism of conduction in conducting polymers with suitable examples. (6 marks)

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